实验要求

Use Wireshark to understand the process of network communication

– Start your device from a clean state, without an IP address
– Run Wireshark to capture packets
– Use your browser to visit a website (e.g., www.baidu.com)
– Read the captured packets in sequence

Requirement

– Capture as more protocol packets as possible, including but not limited to DHCP, ARP, DNS, HTTP, TCP, UDP, STP, …
– A report that describes the detailed communication process, and what you newly understand after this Wireshark lab

实验环境

- **操作系统**：VMware运行的Ubuntu24.04虚拟机
- **工具**：
  - Wireshark（用于捕获和分析网络数据包）
  - firefox（用于访问网站）

实验步骤

1 释放 IP 地址

1. 打开终端，输入以下命令释放 `ens33` 接口的 IP 地址：

   ```
   sudo ip addr flush dev ens33
   ```

2. 确认 IP 地址已被释放：

   ```
   ip addr show
   ```

```
wsp@wsp-VMware-Virtual-Platform:~/桌面$ sudo ip addr flush dev ens33
wsp@wsp-VMware-Virtual-Platform:~/桌面$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
up default qlen 1000
    link/ether 00:0c:29:30:01:70 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
wsp@wsp-VMware-Virtual-Platform:~/桌面$
```

2 启动 Wireshark 捕获

1. 打开 Wireshark。
2. 选择网络接口 `ens33` 。
3. 点击“Start”开始捕获数据包。

3 重启网络接口以重新获取 IP

1. 在终端中输入以下命令重启 `ens33` 接口：

   ```
   sudo ip link set ens33 down
   sudo ip link set ens33 up
   ```

2. 确认 IP 地址已获取：

   ```
   ip addr show
   ```

```
wsp@wsp-VMware-Virtual-Platform:~/桌面$ sudo ip link set ens33 down
wsp@wsp-VMware-Virtual-Platform:~/桌面$ sudo ip link set ens33 up
wsp@wsp-VMware-Virtual-Platform:~/桌面$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
up default qlen 1000
    link/ether 00:0c:29:30:01:70 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.109.128/24 brd 192.168.109.255 scope global dynamic noprefixro
te ens33
       valid_lft 1789sec preferred_lft 1789sec
    inet6 fe80::20c:29ff:fe30:170/64 scope link
       valid_lft forever preferred_lft forever
```

4 访问网站

1. 打开浏览器。
2. 访问网站 `www.bilibili.com` 。
3. 等待页面加载完成。

5 停止捕获并保存数据

返回 Wireshark，点击“Stop”停止捕获。

通信流程分析

1 DHCP 过程

- **DHCP Discover**：
  - 设备广播发送 DHCP Discover 包，寻找可用的 DHCP 服务器。
- **DHCP Offer**：
  - DHCP 服务器回复 DHCP Offer 包，提供一个可用的 IP 地址。
- **DHCP Request**：
  - 设备发送 DHCP Request 包，请求分配 DHCP Offer 中提供的 IP 地址。
- **DHCP Ack**：
  - DHCP 服务器回复 DHCP Ack 包，确认分配 IP 地址。

No.	Time	Source	Destination	Protocol	Length	Info
13	106.025620877	0.0.0.0	255.255.255.255	DHCP	353	DHCP Request - Transaction ID 0xff9756a
14	106.025926715	192.168.109.254	192.168.109.128	DHCP	342	DHCP ACK - Transaction ID 0xff9756a

2 ARP 过程

- **ARP 请求**：
  - 设备发送 ARP 请求包，解析默认网关的 MAC 地址。
- **ARP 响应**：
  - 默认网关回复 ARP 响应包，提供自己的 MAC 地址。

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	VMware_30:01:70	Broadcast	ARP	42	ARP Announcement for 192.168.109.128
6	2.001177696	VMware_30:01:70	Broadcast	ARP	42	ARP Announcement for 192.168.109.128
15	106.028641087	VMware_30:01:70	Broadcast	ARP	42	Who has 192.168.109.128? (ARP Probe)
16	106.070362963	VMware_30:01:70	Broadcast	ARP	42	Who has 192.168.109.128? (ARP Probe)
17	106.109167665	VMware_30:01:70	Broadcast	ARP	42	Who has 192.168.109.128? (ARP Probe)
19	106.169086494	VMware_30:01:70	Broadcast	ARP	42	ARP Announcement for 192.168.109.128
20	106.185577039	VMware_30:01:70	Broadcast	ARP	42	Who has 192.168.109.2? Tell 192.168.109.128
21	106.185781924	VMware_f1:45:78	VMware_30:01:70	ARP	60	192.168.109.2 is at 00:50:56:f1:45:78
52	108.169216404	VMware_30:01:70	Broadcast	ARP	42	ARP Announcement for 192.168.109.128
57	110.169370435	VMware_30:01:70	Broadcast	ARP	42	ARP Announcement for 192.168.109.128

3 DNS 过程

- **DNS 查询**：
  - 设备发送 DNS 查询包，解析 `www.bilibili.com` 的 IP 地址。
- **DNS 响应**：
  - DNS 服务器回复 DNS 响应包，提供目标域名的 IP 地址。

540	121.254475471	192.168.109.2	192.168.109.128	DNS	178	Standard query response 0xc767 HTTPS 427884.sched.sma-dk.tdnsstic1.cn SOA ns1.s…
877	121.448386658	192.168.109.128	192.168.109.2	DNS	77	Standard query 0x5d98 A data.bilibili.com
878	121.448563251	192.168.109.128	192.168.109.2	DNS	77	Standard query 0xa15a HTTPS data.bilibili.com
884	121.462561707	192.168.109.2	192.168.109.128	DNS	281	Standard query response 0x5d98 A data.bilibili.com CNAME data.bilicdn2.com A 11…
885	121.462561871	192.168.109.2	192.168.109.128	DNS	187	Standard query response 0xa15a HTTPS data.bilibili.com CNAME data.bilicdn2.com …
886	121.462939309	192.168.109.128	192.168.109.2	DNS	77	Standard query 0x057e HTTPS data.bilicdn2.com
889	121.472616723	192.168.109.2	192.168.109.128	DNS	150	Standard query response 0x057e HTTPS data.bilicdn2.com SOA ns3.dnsv5.com
900	121.483913056	192.168.109.128	192.168.109.2	DNS	76	Standard query 0xde36 A api.bilibili.com
901	121.484190253	192.168.109.128	192.168.109.2	DNS	76	Standard query 0x791a HTTPS api.bilibili.com
931	121.504051807	192.168.109.2	192.168.109.128	DNS	295	Standard query response 0xde36 A api.bilibili.com CNAME a.w.bilicdn1.com A 119.…
932	121.504051903	192.168.109.2	192.168.109.128	DNS	176	Standard query response 0x791a HTTPS api.bilibili.com CNAME a.w.bilicdn1.com SO…
939	121.504658258	192.168.109.128	192.168.109.2	DNS	76	Standard query 0xfcf8 HTTPS a.w.bilicdn1.com
1096	121.609156016	192.168.109.128	192.168.109.2	DNS	75	Standard query 0x0a5f A cm.bilibili.com
1097	121.609355329	192.168.109.128	192.168.109.2	DNS	75	Standard query 0x3aa4 HTTPS cm.bilibili.com
1115	121.623287075	192.168.109.2	192.168.109.128	DNS	466	Standard query response 0x0a5f A cm.bilibili.com CNAME a.w.bilicdn1.com A 119.8…
1116	121.623287192	192.168.109.2	192.168.109.128	DNS	196	Standard query response 0x3aa4 HTTPS cm.bilibili.com CNAME a.w.bilicdn1.com SOA…

4 mDNS 过程

- **mDNS 查询**：
  - 设备发送 mDNS 查询包，解析本地网络中的服务名称。

- **mDNS 响应**：
  - 本地设备回复 mDNS 响应包，提供服务名称对应的 IP 地址。

56	110.122601885	192.168.109.1	224.0.0.251	MDNS	145	Standard query response 0x0000 PTR, cache flush wsp-VMware-Virtual-Platform.loc…
55	110.122339973	192.168.109.128	224.0.0.251	MDNS	145	Standard query response 0x0000 PTR, cache flush wsp-VMware-Virtual-Platform.loc…
54	109.267128123	192.168.109.1	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "QU" question PTR _ipp._tcp.local, …
53	109.266763319	192.168.109.128	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "QM" question PTR _ipp._tcp.local, …
51	108.053789663	192.168.109.1	224.0.0.251	MDNS	145	Standard query response 0x0000 PTR, cache flush wsp-VMware-Virtual-Platform.loc…
50	108.053486916	192.168.109.128	224.0.0.251	MDNS	145	Standard query response 0x0000 PTR, cache flush wsp-VMware-Virtual-Platform.loc…
47	107.265958840	192.168.109.1	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "QU" question PTR _ipp._tcp.local, …
46	107.265560408	192.168.109.128	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "QM" question PTR _ipp._tcp.local, …
45	106.985577794	192.168.109.1	224.0.0.251	MDNS	145	Standard query response 0x0000 PTR, cache flush wsp-VMware-Virtual-Platform.loc…
44	106.984860198	192.168.109.128	224.0.0.251	MDNS	145	Standard query response 0x0000 PTR, cache flush wsp-VMware-Virtual-Platform.loc…
43	106.785017188	192.168.109.1	224.0.0.251	MDNS	157	Standard query 0x0000 ANY 128.109.168.192.in-addr.arpa, "QU" question ANY wsp-V…
42	106.784793595	192.168.109.128	224.0.0.251	MDNS	157	Standard query 0x0000 ANY 128.109.168.192.in-addr.arpa, "QM" question ANY wsp-V…
36	106.533902025	192.168.109.1	224.0.0.251	MDNS	157	Standard query 0x0000 ANY 128.109.168.192.in-addr.arpa, "QU" question ANY wsp-V…
35	106.533579228	192.168.109.128	224.0.0.251	MDNS	157	Standard query 0x0000 ANY 128.109.168.192.in-addr.arpa, "QM" question ANY wsp-V…
30	106.283494555	192.168.109.1	224.0.0.251	MDNS	157	Standard query 0x0000 ANY 128.109.168.192.in-addr.arpa, "QU" question ANY wsp-V…
29	106.283063092	192.168.109.128	224.0.0.251	MDNS	157	Standard query 0x0000 ANY 128.109.168.192.in-addr.arpa, "QM" question ANY wsp-V…

5 IGMPv3 过程

- **IGMPv3 报告**：
  - 设备发送 IGMPv3 报告包，加入多播组。
- **多播流量**：
  - 设备接收多播流量（如视频流或服务发现流量）。

37	106.782463750	192.168.109.128	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.251 for any sources
18	106.167214460	192.168.109.128	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.251 for any sources

6 TCP

- **SYN**：
  - 设备发送 SYN 包，请求与目标服务器建立 TCP 连接。
- **SYN-ACK**：
  - 目标服务器回复 SYN-ACK 包，确认连接请求。
- **ACK**：
  - 设备发送 ACK 包，完成三次握手。

1600	126.614103312	119.84.174.94	192.168.109.128	TLSv1.2	96	Application Data
1601	126.614152365	192.168.109.128	119.84.174.94	TCP	54	57546 → 443 [ACK] Seq=115977 Ack=20434 Win=65535 Len=0
1602	126.616527260	192.168.109.128	113.142.167.95	TCP	74	35954 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=28134097 TSe
1603	126.633237623	113.142.167.95	192.168.109.128	TCP	60	443 → 35952 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
1604	126.633276253	192.168.109.128	113.142.167.95	TCP	54	35952 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
1605	126.633825314	192.168.109.128	113.142.167.95	TLSv1.2	1962	Client Hello (SNI=impression.biligame.com)
1606	126.634135845	113.142.167.95	192.168.109.128	TCP	60	443 → 35952 [ACK] Seq=1 Ack=1461 Win=64240 Len=0
1607	126.634135943	113.142.167.95	192.168.109.128	TCP	60	443 → 35952 [ACK] Seq=1 Ack=1909 Win=64240 Len=0
1608	126.641952899	113.142.167.95	192.168.109.128	TCP	60	443 → 35954 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
1609	126.642007352	192.168.109.128	113.142.167.95	TCP	54	35954 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
1610	126.642742909	192.168.109.128	113.142.167.95	TLSv1.2	1962	Client Hello (SNI=impression.biligame.com)
1611	126.642910877	113.142.167.95	192.168.109.128	TCP	60	443 → 35954 [ACK] Seq=1 Ack=1461 Win=64240 Len=0
1612	126.642910960	113.142.167.95	192.168.109.128	TCP	60	443 → 35954 [ACK] Seq=1 Ack=1909 Win=64240 Len=0
1613	126.661224961	119.84.174.67	192.168.109.128	TLSv1.2	808	Application Data
1614	126.661250651	192.168.109.128	119.84.174.67	TCP	54	50912 → 443 [ACK] Seq=15057 Ack=15132 Win=65535 Len=0
1615	126.684318232	119.84.174.94	192.168.109.128	TLSv1.2	492	Application Data, Application Data

7 TLSv1.2/v1.3 握手

- **TLSv1.2 握手**：
  - 设备与服务器通过 2-RTT 握手建立 TLS 连接。
- **TLSv1.3 握手**：
  - 设备与服务器通过 1-RTT 握手建立 TLS 连接。

1779	128.109039026	192.168.109.128	113.142.186.225	TLSv1.3	78	Application Data
1780	128.109105967	192.168.109.128	113.142.186.225	TCP	54	57686 → 443 [FIN, ACK] Seq=2722 Ack=65748 Win=65535 Len=0
1781	128.109312359	113.142.186.225	192.168.109.128	TCP	60	443 → 57686 [ACK] Seq=65748 Ack=2722 Win=64240 Len=0
1782	128.109312458	113.142.186.225	192.168.109.128	TCP	60	443 → 57686 [ACK] Seq=65748 Ack=2723 Win=64239 Len=0
1783	128.109491426	192.168.109.128	61.159.93.100	TCP	54	44708 → 80 [FIN, ACK] Seq=1001 Ack=3779 Win=60462 Len=0
1784	128.109562505	192.168.109.128	202.110.100.241	TLSv1.3	78	Application Data
1785	128.109601461	192.168.109.128	202.110.100.241	TCP	54	35280 → 443 [FIN, ACK] Seq=6533 Ack=281304 Win=65535 Len=0
1786	128.109655532	61.159.93.100	192.168.109.128	TCP	60	80 → 44708 [ACK] Seq=3779 Ack=1002 Win=64239 Len=0
1787	128.109655630	202.110.100.241	192.168.109.128	TCP	60	443 → 35280 [ACK] Seq=281304 Ack=6533 Win=64240 Len=0
1788	128.109716850	202.110.100.241	192.168.109.128	TCP	60	443 → 35280 [ACK] Seq=281304 Ack=6534 Win=64239 Len=0
1789	128.109837037	192.168.109.128	119.84.174.94	TLSv1.2	85	Encrypted Alert
1790	128.109900091	192.168.109.128	119.84.174.94	TCP	54	57546 → 443 [FIN, ACK] Seq=125205 Ack=24232 Win=65535 Len=0
1791	128.109940149	119.84.174.94	192.168.109.128	TCP	60	443 → 57546 [ACK] Seq=24232 Ack=125205 Win=64240 Len=0
1792	128.110016209	119.84.174.94	192.168.109.128	TCP	60	443 → 57546 [ACK] Seq=24232 Ack=125206 Win=64239 Len=0

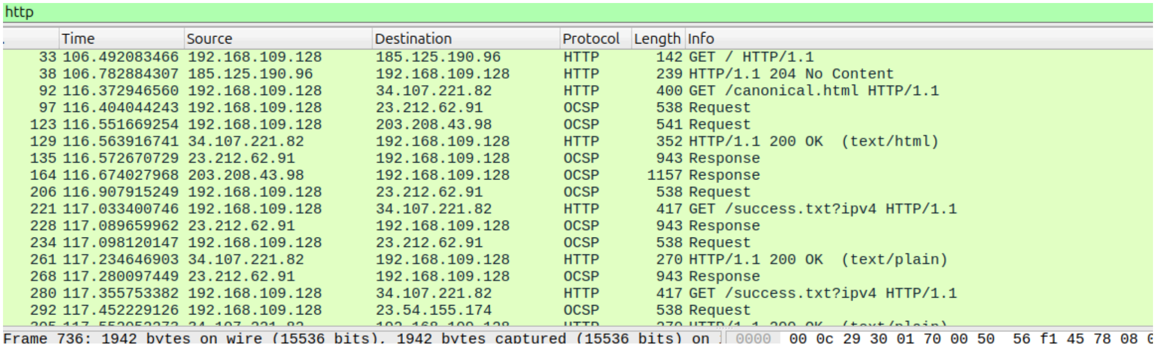
8 OCSP 过程

- **OCSP 请求**：

    - 设备向 OCSP 服务器发送请求，检查服务器证书的状态。
- **OCSP 响应**：
    - OCSP 服务器回复响应，指示证书的状态。

9 HTTP 过程

- **HTTP 请求**：
    - 设备发送 HTTP GET 请求，请求网页内容。
- **HTTP 响应**：
    - 目标服务器回复 HTTP 响应包，包含网页内容。


http

	Time	Source	Destination	Protocol	Length	Info
33	106.492083466	192.168.109.128	185.125.190.96	HTTP	142	GET / HTTP/1.1
38	106.782884307	185.125.190.96	192.168.109.128	HTTP	239	HTTP/1.1 204 No Content
92	116.372946560	192.168.109.128	34.107.221.82	HTTP	400	GET /canonical.html HTTP/1.1
97	116.404044243	192.168.109.128	23.212.62.91	OCSP	538	Request
123	116.551669254	192.168.109.128	203.208.43.98	OCSP	541	Request
129	116.563916741	34.107.221.82	192.168.109.128	HTTP	352	HTTP/1.1 200 OK (text/html)
135	116.572670729	23.212.62.91	192.168.109.128	OCSP	943	Response
164	116.674027968	203.208.43.98	192.168.109.128	OCSP	1157	Response
206	116.907915249	192.168.109.128	23.212.62.91	OCSP	538	Request
221	117.033400746	192.168.109.128	34.107.221.82	HTTP	417	GET /success.txt?ipv4 HTTP/1.1
228	117.089659962	23.212.62.91	192.168.109.128	OCSP	943	Response
234	117.098120147	192.168.109.128	23.212.62.91	OCSP	538	Request
261	117.234646903	34.107.221.82	192.168.109.128	HTTP	270	HTTP/1.1 200 OK (text/plain)
268	117.280097449	23.212.62.91	192.168.109.128	OCSP	943	Response
280	117.355753382	192.168.109.128	34.107.221.82	HTTP	417	GET /success.txt?ipv4 HTTP/1.1
292	117.452229126	192.168.109.128	23.54.155.174	OCSP	538	Request

Frame 736: 1942 bytes on wire (15536 bits), 1942 bytes captured (15536 bits) on | 0000 00 0c 29 30 01 70 00 50 56 f1 45 78 08

4. 实验总结

在本次实验中，我成功地观察到了设备从没有 IP 地址到访问网站的通信过程，涉及到了以下协议：

1. **DHCP**：设备如何获取 IP 地址。
2. **ARP**：设备如何解析默认网关的 MAC 地址。
3. **DNS/mDNS**：设备如何解析域名或本地服务名称。
4. **IGMPv3**：设备如何加入多播组。
5. **TCP**：设备如何与目标服务器建立连接。
6. **TLSv1.2/v1.3**：设备如何与目标服务器建立加密连接。
7. **OCSP**：设备如何检查服务器证书的状态。
8. **HTTP**：设备如何请求和接收网页内容。

我在这次wireshark lab后的新认识：

通过本次 Wireshark 实验，我对网络通信协议有了更深入的理解，尽管我已在计算机网络课程中学习过 ARP、DNS、TCP/IP 和 HTTP 等协议。
首先是了解到了一个新的协议，DHCP协议，这是在此之前我未接触过的，了解了设备获取IP地址的通信流程。
通过捕获和分析实际的数据包，我看到了这些协议在真实场景中的交互过程。例如，我观察到了 TCP 三次握手的详细步骤。此外，我还了解了 OCSP 如何通过实时检查证书状态来增强 HTTPS 的安全性。通过使用 Wireshark，我还掌握了数据包过滤和分析的实用技能。